

CheXWorld: Exploring Image World Modeling for Radiograph Representation Learning

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CheXWorld: Exploring Image World Modeling for Radiograph Representation Learning

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Motivation: Building World Models

- Humans rely on internal **world models** to understand their environment—enabling reasoning, prediction, and adaptation with minimal supervision. These models encode **common sense knowledge** of the world.
- JEPA (Joint Embedding Predictive Architecture)** offers a compelling framework for implementing such models in machine learning.

Our Goal: To build a **self-supervised model** that captures **essential medical knowledge**, much like a radiologist's internal understanding of radiographic images.

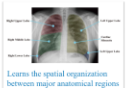
Approach: Three Radiology-Inspired Tasks

Local Anatomical Structure Modeling



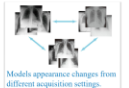
Captures fine-grained details of bones and anatomical structures.

Global Anatomical Layout Modeling



Learns the spatial organization between major anatomical regions.

Domain Variation Modeling



Models appearance changes from different acquisition settings.

Context x: Image with masked regions removed

Target y: masked regions

Latent z: Spatial indices of the masked patches $\mathbf{M}$

Context x: Image crop from one anatomical region

Target y: Image crop from a different anatomical region

Latent z: Relative spatial displacement $\Delta_{x \rightarrow y}$

Context x: Augmented version of the original image

Target y: Original image

Latent z: Augmentation parameters $\mathbf{a}$

JEPA

(Joint Embedding Predictive Architecture)

for Medical Image

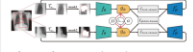
Experiments: Predictive Capability

Visualization of Local Anatomical Structure Modeling (Obtained via RCDM)

Visualization of Global Anatomical Layout Modeling (Obtained via RCDM)

Experiments: Downstream Transfer

The Unified Framework



Design unified contexts, targets, and latent variables to perform the three world modeling tasks simultaneously.

$$\mathcal{L}_{1 \rightarrow 1} = \mathcal{L}_{\text{local} + \text{domain}}(x_1; y_1)$$

$$= \sum \|g_{\theta_1}(h_{x_1}; M_1; a_1) - f_{\theta_1}(y_1)\|_2^2$$

$$\mathcal{L}_{1 \rightarrow 2} = \mathcal{L}_{\text{global} + \text{domain}}(x_1; y_2)$$

$$= \|g_{\theta_1}(h_{x_1}; \Delta_{x_1 \rightarrow y_2}; a_1) - f_{\theta_1}(y_2)\|_2^2$$

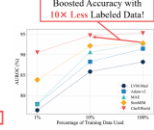
Disease Classification Results

Consistent improvements over existing methods.

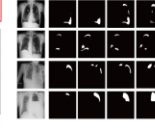
Method	Valid-C100	Validation	Check-ep14	B3VA-C100	Ch-Spot
Scratch	76.22±1.95	82.24±0.80	71.69±0.52	66.99±0.39	80.78±0.03
McG+V1[14]	87.28±0.63	92.68±1.00	79.20±0.30	72.79±0.52	87.12±0.36
DDP[13]	82.89±1.59	90.94±0.29	78.25±0.47	71.27±0.45	87.66±0.62
BEST[1]	85.81±1.98	92.87±1.08	79.91±0.24	72.78±0.37	87.77±0.38
LVA-Med[14]	88.22±0.44	94.81±1.32	80.06±0.09	72.75±0.44	88.87±0.25
MAE[11,15]	92.76±0.18	97.63±0.21	83.67	73.75±0.24	89.37
StdMM[11,16]	92.81±0.31	98.09±0.13	83.64±0.15	74.09±0.39	89.14±0.22
Adam-v2[17]	91.46±0.33	97.807	83.87	73.40±0.08	88.90±0.36
CheXWorld	95.34±0.13	98.68±0.06	83.93±0.03	75.03±0.29	89.63±0.13

Low-data Results

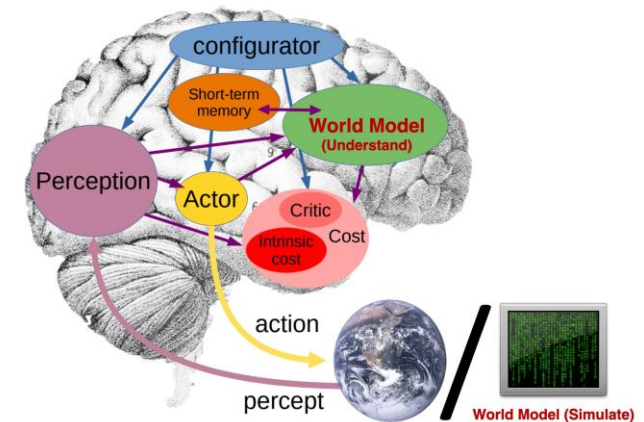
Boosted Accuracy with 0.5% Less Labeled Data!



Segmentation Results

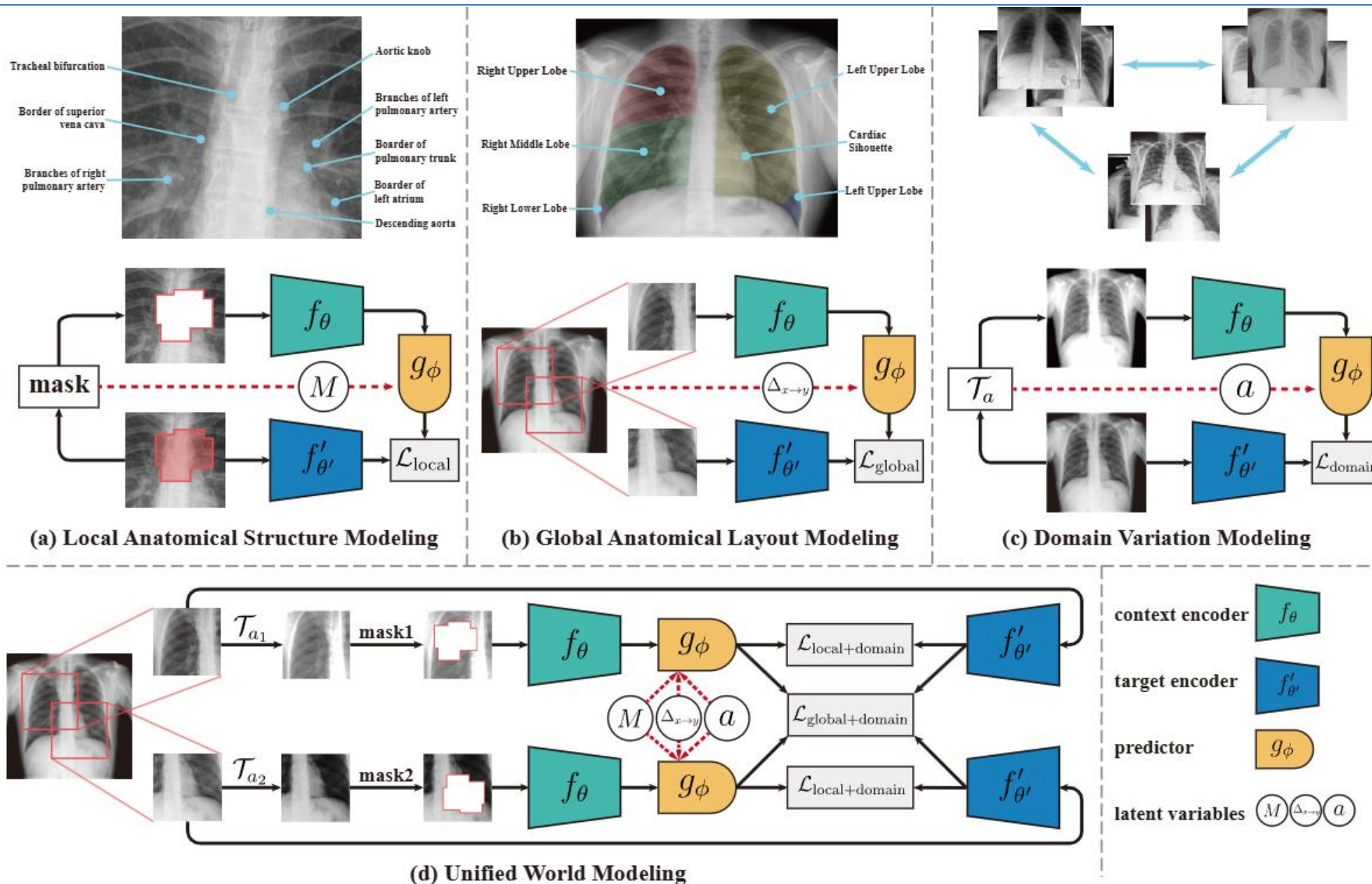


- Introduction
- Related Work
- Basic Framework of World Modeling
- CheXWorld: World Modeling for Radiograph Representation Learning
- Experiments
- Conclusion



- 인간은 **common sense knowledge**를 인코딩하여 세상이 어떻게 작동하는지 알려주고 행동의 결과를 예측하는 internal world model을 개발 할 수 있음.
 - 세상의 역학, 무엇이 가능하고 불가능한지, 그들의 행동의 결과는 어떤지 ...
- Pre-trained world model은 의미론적으로 풍부한 임베딩을 가지고 있기 때문에 (1) 제한된 데이터에서 (2) 다양한 task를 수행할 수 있음.
- Medical domain도 다른 분야 처럼 task-specific models 에서 범용 foundation model을 개발하는 방향으로 연구되고 있음.
 - 이를 위해 의미있는 의료 지식을 인코딩하길 기대하며 Massive data 위에서 사전학습함.

- human anatomy에 깊은 이해를 갖춘 radiologist는 anatomical structures의 차이와 anatomical regions 간의 상대적인 기하학적 관계를 파악할 수 있음.
- 따라서 human anatomy의 계층 구조를 따라 아래 세가지 modeling task를 수행함.
 - local anatomical structure modeling
 - Localized anatomical region 내의 특징과 구조 (e.g., 혈관, 뼈의 모양 ...)
 - global anatomical layout modeling
 - 전반적인 모양과 공간적인 관계 (e.g., 심장과 폐의 모양과 심장은 폐 사이에 있는 것 ...)
 - domain variation modeling
 - Domain간 예측 가능한 transition을 인코딩하는 표현 공간 예측 (e.g., 서로 다른 vendor의 CXR ...)
- 물론 모든 영역을 커버하는 modeling은 아닐지라고 radiograph 판독에 fundamenta한 요소이며 비슷한 조건의 SSL baseline들 보다 일관되게 성능이 좋음을 보임.

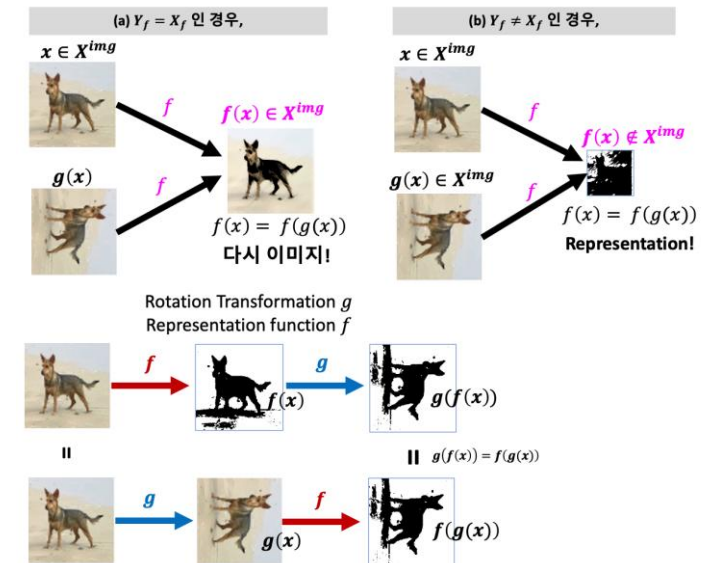


- World Modeling

- model predictive control에서 기원했으며 보통 RL에서 이루어짐.
- 최근에는 common sense knowledge를 잘 알고 있는 world model을 만드는 것이 AGI로 가는 중요한 step으로 믿어지고 있음.
- Input space를 예측하는 영상 생성 모델이 있는 반면 (Sora, VideoPoet),
- **latent space**를 예측하는 것이 human brain 속에 내제된 world model에 더 가깝다고 보여짐.

- Self-supervised learning in medical imaging

- **Restorative:** MAE 계열 (random masking, pixel shuffling 등)
- **Contrastive:** global/dense contrastive learning
- ChexWorld는 최초로 world modeling 개념을 의료 SSL에 도입함.
- contrastive 기반 Adam과 달리 **equivariant representation** 사용
 - invariant: 변환해도 임베딩이 같음 → 정보 손실 가능
 - equivariant: 변환하면 임베딩도 예측 가능한 방식으로 변함 → 정보 보존



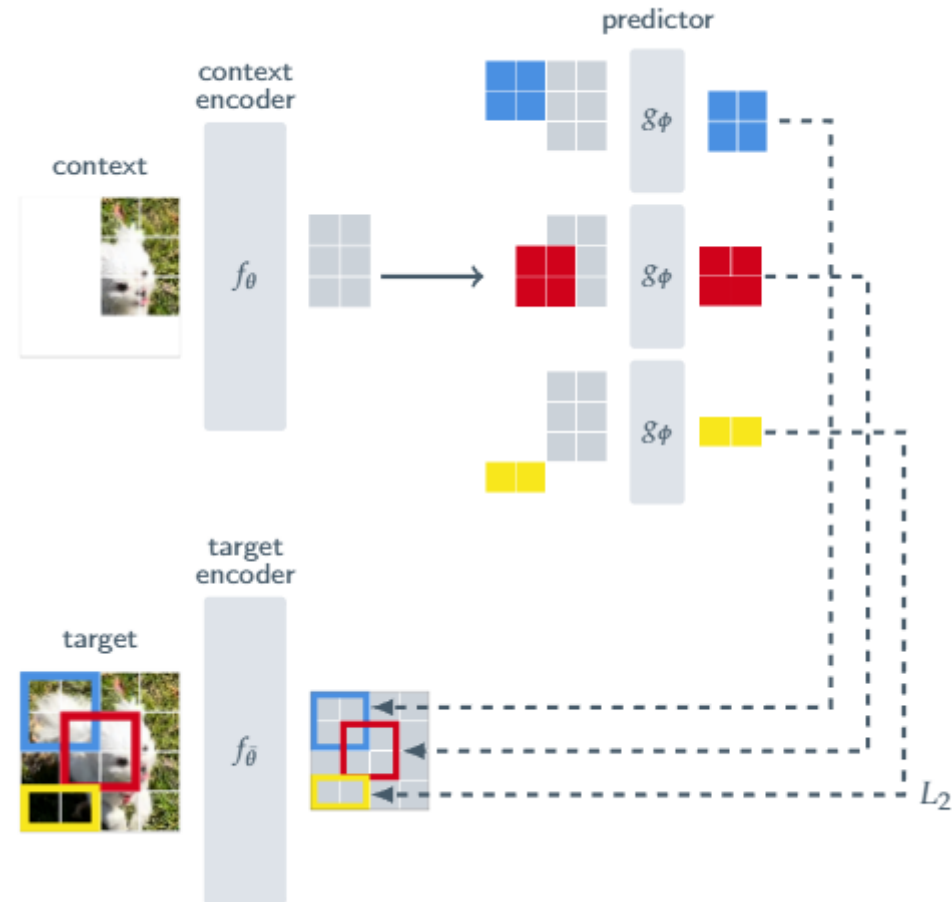
Definition 1.1 (g -invariant). For given f and g , f is g -invariant if for all x ,

$$f(g(x)) = f(x) \quad (1)$$

Definition 1.2 (g -equivariant). For given f and g , f is g -equivariant if for all x ,

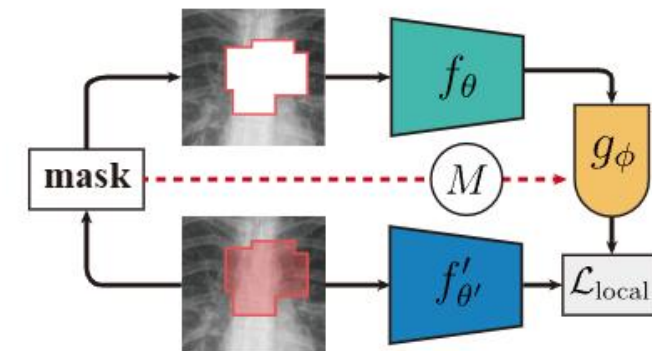
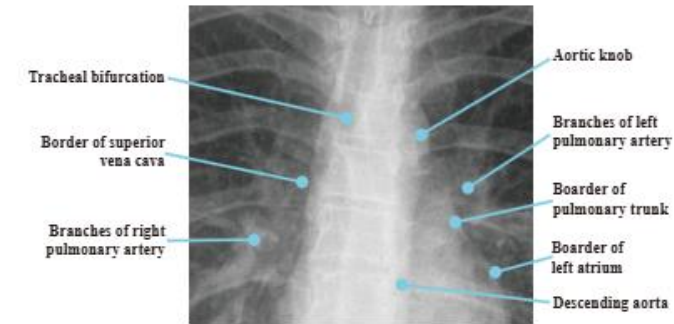
$$f(g(x)) = g(f(x)) \quad (2)$$

Basic Framework of World Modeling



- Local Anatomical Structure Modeling

- 상대적으로 낮은 fundamental level에서 다양한 유형의 tissue(e.g., bones, muscles ...)가 human body의 structural organization을 형성함.
- 이 micro structure의 fine-grained characteristics (e.g., shapes, sizes, appearance, and textures)을 이해하는 것이 Radiologist의 핵심 역량임.
- I-JEPA 수행 (0.15~0.2의 multi block)



(a) Local Anatomical Structure Modeling

- Global Anatomical Layout Modeling

- 인체 내부에서 Tissue group이나 Organ system이 어떻게 anatomically 구성되어 있는지 아는 것도 중요함.
- 주어진 context 밖의 영역을 예측하도록 하여 인체 내의 long range topology를 관계를 학습 시킴.
- 따라서 이는 Local Anatomical Structure Modeling과 orthogonal함.

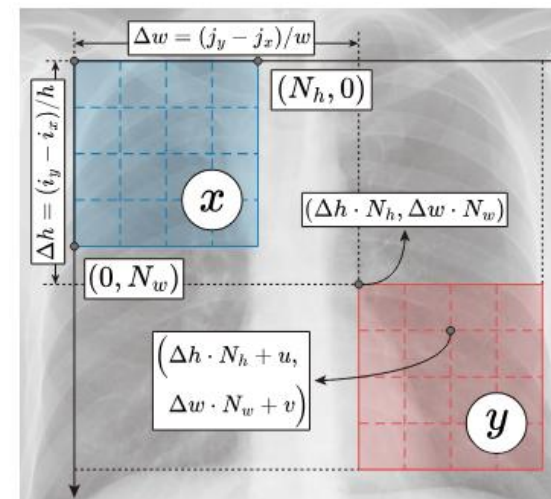
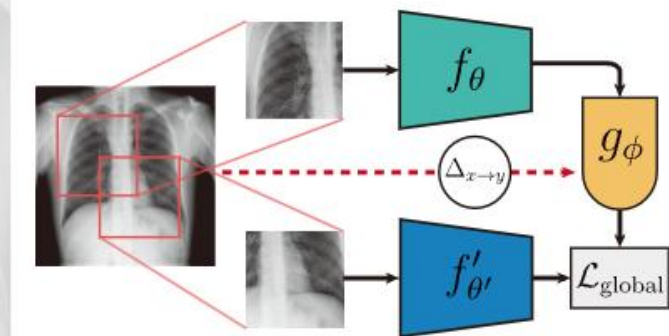
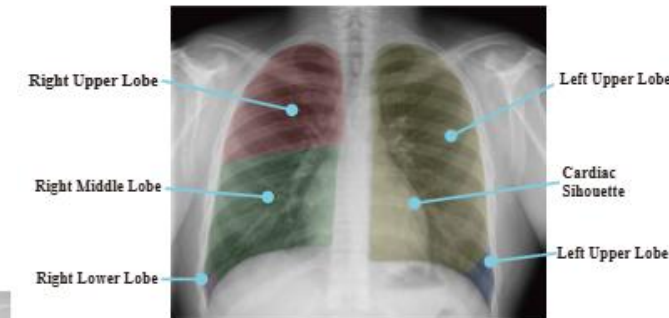


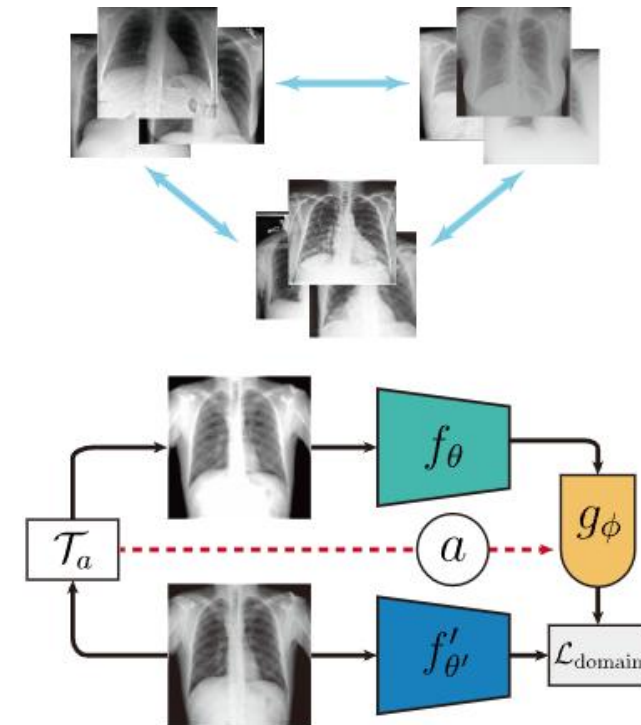
Figure 3. Formulation of global anatomical layout modeling.



(b) Global Anatomical Layout Modeling

- Domain Variation Modeling

- Radiologist는 human body의 anatomical knowledg의 두 단계 뿐 아니라 이를 넘어서 각 radiograph의 특성에 유연하게 adaptin하며 약간의 변화에도 이를 이해할 수 있어야 함.
 - from different sources (e.g., hospitals, techniques, equipment)
 - show diverse appearances (e.g., clarity, contrast, exposure).
- Radiologist에 내재된 world model이 domain 변화를 시뮬레이션 하여 scanned body에 대한 객관적인 view를 얻을 수 있다고 가정함.
- Gaussian blur, brightness, contrast, and gamma adjustments를 이 용해서 domain shifts를 시뮬레이션함.
 - Equivariant하여 representation 예측 가능한 방식으로 학습됨.



(c) Domain Variation Modeling

- The Unified Framework

- 3개의 Task를 한번에 수행하기 위해 2개의 crop을 이용해서 Local, global을 수행함.
- augmented and masked with two different configurations: $\{(a_1, M_1), (a_2, M_2)\}$

$$x_i = \text{Mask}(T_{a_i}(y_i), M_i), i = 1, 2.$$

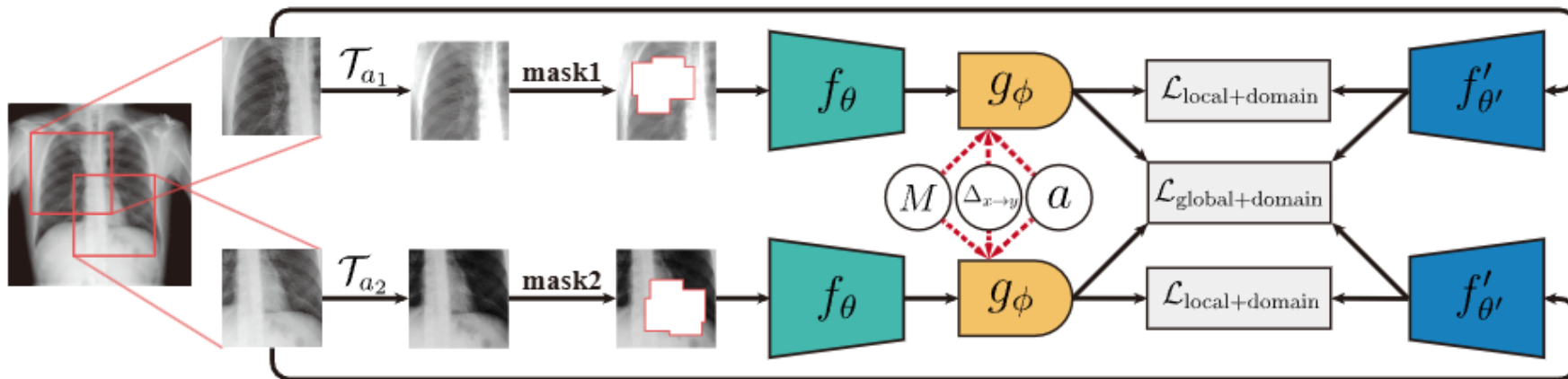
$h_{x_1}, h_{x_2}, h_{y_1}, h_{y_2}$: each representation

$\Delta_{x_1 \rightarrow y_2}$: relative position information

$$\mathcal{L}_{1 \rightarrow 1} = \mathcal{L}_{\text{local}+\text{domain}}(x_1, y_1) = \sum_{c \in M_1} \|g_\phi(h_{x_1}; M_1, a_1)_c - f'_\theta(y_1)\|_2^2$$

$$\mathcal{L}_{1 \rightarrow 2} = \mathcal{L}_{\text{global}+\text{domain}}(x_1, y_2) = \|g_\phi(h_{x_1}; \Delta_{x_1 \rightarrow y_2}, a_1) - f'_\theta(y_2)\|_2^2$$

$$\mathcal{L} = \mathcal{L}_{1 \rightarrow 1} + \mathcal{L}_{2 \rightarrow 2} + \mathcal{L}_{1 \rightarrow 2} + \mathcal{L}_{2 \rightarrow 1}$$



(d) Unified World Modeling

- Dataset:
 - MIMIC-CXR + NIH ChestX-ray14 + CheXpert, frontal scan ~0.5M radiographs
 - NIH / CheXpert의 val/test split은 제외 (data leakage 방지)
- Backbone:
 - Context encoder: ViT-Base, patch 16×16
 - Target encoder: Context encoder의 EMA (초기 ratio 0.996 → 1.0, cosine schedule)
 - Predictor: 6-layer, 384-dim embedding
 - Positional encoding: sinusoidal
- Train:
 - Optimizer: AdamW ($\beta_1=0.9$, $\beta_2=0.999$)
 - LR: 2×10^{-4} , weight decay 0.05, gradient clipping 1.0
 - Schedule: linear warmup 40 epoch → cosine annealing
 - Loss: raw predictor output vs. **layer-normalized** target encoder output의 L2 (MSE)
 - 300 epochs, batch 2048, 8× RTX 4090 (24GB), 16시간

CheXWorld is a Strong World Predictor

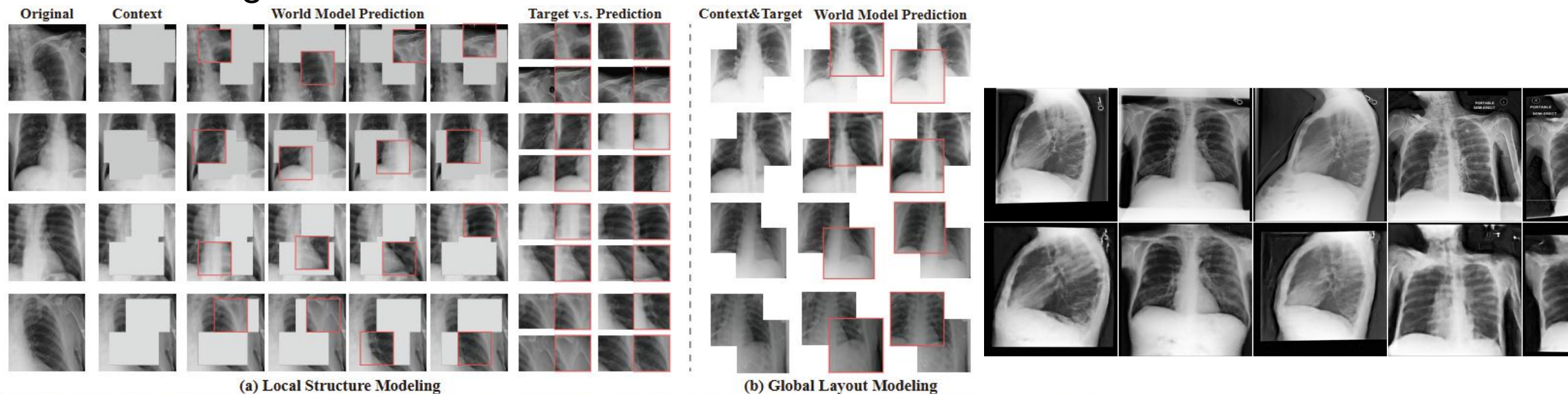


Figure 4. **Visualization of the CheXWorld predictor outputs** (zooming in for details). The images presented in this figure were not included in the pre-training of CheXWorld or the training of the diffusion model. Regions in red bounding boxes denote the predictor outputs that are mapped to pixel space using the RCDM [8] framework. In (a), gray areas indicate masked regions excluded from the context. In (b), the two overlapping regions alternately serve as context and target.

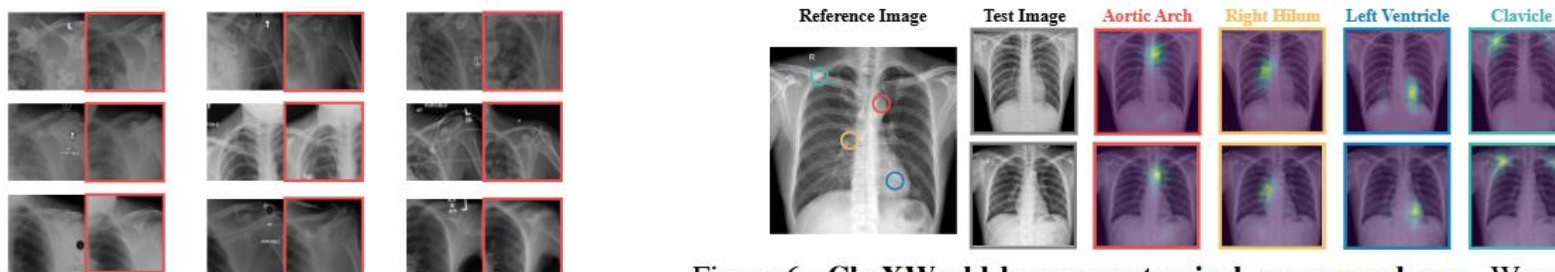


Figure 5. **CheXWorld prioritizes relevant medical features over spurious signals** (e.g., lateral markers) in the image. Regions in red boxes denote the predictor outputs.

Figure 6. **CheXWorld learns anatomical correspondence.** We compute pixel-level embeddings using RoI-pooling [23] and calculate the embedding similarity between four anatomical landmarks in a reference image and each pixel in the test image to create feature similarity heatmaps.

CheXWorld Produces Transferable Representations

Table 1. Performance on five downstream classification benchmarks. Accuracy is reported for RSNA, and the area under the ROC curve (AUROC) is reported for the rest of the datasets. The best two results are **bold-faced** and underlined, respectively.

Method	Pre-train Data	Backbone	VinDr-CXR	ShenZhen	ChestX-ray14	RSNA (CLS)	CheXpert
Scratch	-	ViT-B	70.22±1.95	82.24±0.60	71.69±0.32	66.59±0.39	80.78±0.03
MoCo-v3 [14]	IN-1k (1.3M)	ViT-B	87.25±0.63	92.85±1.00	79.20±0.30	72.79±0.52	87.12±0.36
DINO [11]	IN-1k (1.3M)	ViT-B	82.89±1.10	90.39±4.29	78.37±0.47	71.27±0.45	87.01±0.62
BEiT [6]	IN-21k (12M)	ViT-B	85.93±1.98	92.87±1.08	79.91±0.24	72.78±0.37	87.77±0.38
LVM-Med [48]	Medical (1.3M)	ViT-B	88.22±0.44	94.81±1.32	80.08±0.09	72.75±0.44	88.07±0.25
MAE [32, 67]	X-rays (0.5M)	ViT-B	92.76±0.18	97.63±0.21	83.0 [†]	73.75±0.24	<u>89.3[†]</u>
SimMIM [46, 68]	IN-21k (12M) & X-rays (0.9M)	Swin-B	<u>92.81±0.31</u>	<u>98.09±0.13</u>	83.04±0.15	<u>74.09±0.39</u>	89.14±0.22
Adam-v2 [33]	IN-21k (12M) & X-rays (0.9M)	ConvNeXt-B	91.46±0.33	97.80 [†]	<u>83.4[†]</u>	73.40±0.88	88.90±0.36
CheXWorld	X-rays (0.5M)	ViT-B	95.24±0.13	98.88±0.06	83.58±0.05	75.03±0.39	89.63±0.13
Rad-DINO [‡] [55]	LVD (142M) & X-rays (public+private)	ViT-B	95.16±0.16	98.20±0.17	83.61±0.08	74.51±0.46	88.94±0.15

Results on ImageNet pre-trained models are adopted from [46]. [†]Results reported by the original authors [32, 33]. [‡]Rad-DINO [55] is included for reference but not for direct comparison, as it requires 2560 GPU-hours (20 times more than ours) and is trained on both public and private datasets.

CheXWorld Produces Transferable Representations

Table 2. Results on segmentation (left) and few-shot learning (right) tasks. The dice score and the AUROC score are reported for the segmentation and few-shot learning benchmarks respectively.

Method	Segmentation		Few-shot (MedFMC-ChestDR)		
	SIIM-ACR	RSNA (SEG)	1-shot	5-shot	10-shot
LVM-Med [48]	82.19±0.30	78.47±0.24	57.55±0.81	66.95±0.60	67.44±0.71
MAE [32, 67]	83.01±0.39	77.39±0.25	61.31±0.27	74.03±0.30	75.26±0.36
SimMIM [46, 68]	82.89±0.07	78.31±0.41	60.64±1.92	72.14±0.54	74.42±0.38
Adam-v2 [33]	83.60±0.33	78.53±0.19	59.16±1.03	70.28±0.39	70.67±0.71
CheXWorld	84.58±0.34	79.02±0.25	64.60±1.00	75.19±0.51	76.40±0.25

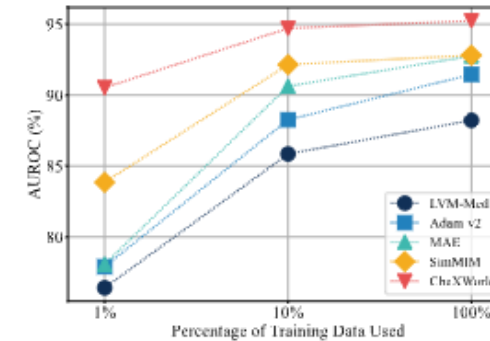


Figure 7. Fine-tuning with 1%, 10%, and 100% training data on VinDr-CXR.

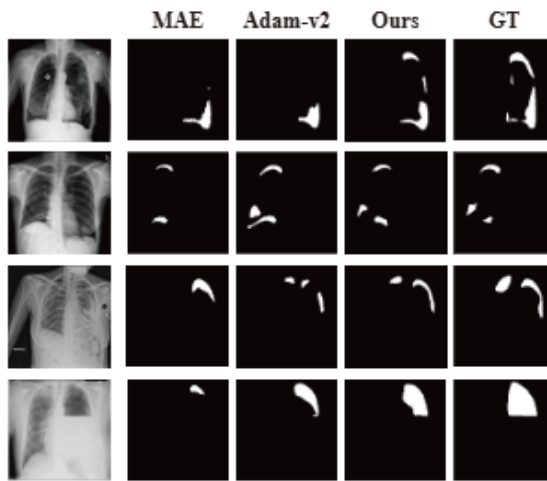


Figure 8. Visualization of segmentation masks on SIIM-ACR pneumothorax dataset. GT stands for the ground truth masks.

Table 3. Ablation studies of the world modeling tasks and latent variables. We report the results on VinDr-CXR and RSNA with 1% proportion of training data. AUROC and accuracy are reported for the respective datasets. “Local,” “Global,” and “Domain.” are abbreviations for local anatomical structure modeling, global anatomical layout modeling, and domain variation modeling, respectively. “Unified” indicates whether our unified formulation is applied, as opposed to a naive combination of losses.

World Modeling Task			Latent Variables		Unified	VinDr 1%	RSNA 1%
Local.	Global.	Domain.	$\Delta_{x \rightarrow y}$	a			
✓					N.A.	84.71±1.00	65.89±0.54
✓	✓		✓		✓	88.31±0.88	67.47±0.51
✓		✓		✓	✓	89.73±0.50	68.27±1.00
✓	✓	✓	✓	✓	✓	90.53±1.01	70.01±0.33
✓	✓	✓	✓	✓		89.12±0.76	68.37±0.50
✓	✓	✓	✓		✓	89.91±0.60	68.98±0.60
✓	✓	✓		✓	✓	87.60±0.72	66.56±0.39

- CheXWorld는 CXR을 위한 **self-supervised world modeling framework**로, local, global, domain variation의 세 가지를 동시에 학습함.
- 세 task를 단일 pipeline으로 통합하여 8개 benchmark에서 SOTA를 달성함.
- general-purpose foundation model for medical vision을 향한 new insight를 제공함.